

1) Grey Level Transformation.

$$\text{Transformation } s = 255 - r$$

$$\text{Pixel } r = 50, 100, 150, 200$$

(a) purpose of grey-level transformation

Grey-level transformation is used to modify the intensity value of pixels in an image to improve its visual appearance or make important feature easier to analyze.

(b) Negative transformation:

$$s = 255 - r$$

* r = original pixel

* s = transformed pixel

* 255 = Max

original r	Calculation	s
50	$255 - 50$	205
100	$255 - 100$	155
150	$255 - 150$	105
200	$255 - 200$	55

$$s = 205, 155, 105, 55,$$

2) Histogram Equalization

Given:

Intensity x_k	Frequency n_k
0	4
1	6
2	10
3	6

Total number of Pixels:

$$N = 4 + 6 + 10 + 6 = 26$$

Since 4-bit Image:

$$L = 2^4 \\ = 16$$

a) Why Histogram equalization is used:

Histogram equalization is used to improve image contrast by redistributing the intensity values over a wider range.

- + Enhance poorly image.
- + makes details more visible
- + Produces a more uniform intensity distribution

(b) Cumulative Distribution Function (CDF):

Intensity	Frequency	Cumulative Frequency	CDF
0	1	1	$1/26 = 0.1538$
1	6	7	$7/26 = 0.2692$
2	10	17	$17/26 = 0.6538$
3	6	23	$23/26 = 0.8846$

$$CDF = [0.1538, 0.2692, 0.6538, 0.8846]$$

Ex) Equalized Intensity values:

Formula: $S_k = \text{round} \left[(L-1) CDF(x_k) \right]$

$$L-1 = 15$$

$$\ln 0: S_0 = 15(0.1538) = 2.307$$

$$\boxed{S_0 = 2}$$

$$\ln 1: S_1 = 15(0.2692) = 4.038$$

$$\boxed{S_1 = 4}$$

$$\ln 2: S_2 = 15(0.6538) = 9.807$$

$$\boxed{S_2 = 9}$$

$$\ln 3: S_3 = 15(0.8846) = 13.269$$

$$\boxed{S_3 = 13}$$

1) Gaussian Smoothing:

(a) Why Gaussian? Giving a preferred view on image.

Why?

Gaussian filtering gives more importance to pixels near the center and less importance to pixels further away.

Gaussian kernel:

$$G = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

(b) Calculation the Center Pixel.

Input Image:

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Gaussian kernel:

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\begin{aligned} \text{New Pixel} &= \frac{1}{16} [C_1 \times I_{10} + C_2 \times I_{20} + C_3 \times I_{10} \\ &\quad + C_4 \times I_{20} + C_5 \times I_{40} + C_6 \times I_{20} + C_7 \times I_{10} \\ &\quad + C_8 \times I_{20} + C_9 \times I_{10}] \end{aligned}$$

$$= \frac{1}{16} [10 + 20 + 10 + 20 + 160 + 20 + 10 + 20 + 10]$$

24/10/16
New Combo Price = 22.9

5) Edge Detection Using Sobel Operator:

(a) Concept and purpose of edge detection:

Edge detection is an image-processing technique used to identify locations where there is a significant change in pixel intensity.

- + object boundaries
- + Changes in Surface
- + Shape Information
- + Structural details

(b) Sobel Calculation:

Horizontal Sobel mask:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 16 & 20 & 30 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

$$G_{20} = (-1)(10) + (-2)(10) + (-1)(10) + (0)(00) + (0)(20) \\ + (0)(30) + (1)(20) + (0)(30) + (1)(20)$$

$$= -10 - 20 - 10 + 0 + 0 + 0 + 30 + 60 + 30$$

$$= 40 + 120$$

$$\boxed{G_{20} = 80}$$

5) Contrast Stretching:

(a) Need for Contrast Stretching:

Contrast stretching is used when an image occupies only a small portion of the available intensity range.

- * Expands the intensity range.
- * Improves visibility of details.
- * Increases image contrast.

(b) Linear Contrast Stretching:

Formula:

$$g = \frac{(b - a_{min})(s_{max} - s_{min})}{s_{max} - s_{min}}$$

Given:

$$d_{\min} = 50, d_{\max} = 150, S_{\min} = 0$$

$$S_{\max} = 255$$

$$S = \frac{(d-50)(255-0)}{150-50} = \frac{255(d-50)}{100}$$

For $d = 50$:

$$S = \frac{255(50-50)}{100}$$

$$\boxed{S = 0}$$

For $d = 100$:

$$S = \frac{255(100-50)}{100} = \frac{255(50)}{100}$$

$$\boxed{S = 127.5 \approx 128}$$

For $d = 150$:

$$S = \frac{255(150-50)}{100} = \frac{255(100)}{100}$$

$$\boxed{S = 255}$$

$\therefore 50 \rightarrow 0, 100 \rightarrow 128, 150 \rightarrow 255,$